

Graduation Project Report

BSc Project

Department of Computer Information Systems

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Dubljly: AI-Powered Video Dubbing and Translation Platform

**Enhancing Educational Access for Arabic-Speaking
Students**

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Abstract

Dublily is an AI-powered platform designed to enhance educational access for Arabic-speaking students by transforming English educational videos into Arabic-dubbed content. The system integrates advanced speech-to-text (Fast-Whisper), large language models (Groq API, Ollama), and text-to-speech (XTTS-v2) technologies to provide natural Arabic dubbing, preserve technical terminology, and offer interactive features such as summaries, keywords, and a chatbot for question answering. This report documents the project's development, from system analysis to implementation, validating its effectiveness in addressing language barriers in education. Targeting Jordanian students and institutions, Dublily aims to democratize access to global educational content, with plans for expansion across the Arab world.

Keywords: Video Dubbing, AI Translation, Arabic Education, Speech Processing

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1 Introduction

1.1 Purpose of the Project

Dubljlly addresses a critical educational barrier affecting millions of Arabic-speaking students: limited access to high-quality educational resources due to language constraints. While the vast majority of premium academic content is produced in English, many students in the Arab world lack the proficiency needed to fully comprehend complex concepts, significantly hindering their educational progress and career opportunities [1]. In Jordan, only 13% of students possess sufficient English proficiency to access advanced academic material, despite English being taught from early grades [4].

This digital divide perpetuates educational inequality, as students without strong English skills cannot access the same wealth of knowledge as their English-proficient peers. Traditional solutions like subtitles or text translations often fail to create immersive learning experiences, while professional dubbing services are prohibitively expensive, costing \$2,000–\$8,000 per hour, and time-consuming, requiring 4–8 weeks per hour of content [1]. Dubljlly harnesses advanced AI technologies to bridge this gap through an innovative platform that transforms English educational videos into accessible Arabic content while preserving critical technical terminology.

The Dubljlly solution:

- Employs cutting-edge speech recognition (FastWhisper) and machine translation (fine-tuned large language models via Groq API) to generate natural-sounding Arabic dubbing that maintains the educational integrity of the original content.
- Preserves essential technical terms in English (e.g., “algorithm,” “photosynthesis”) while providing contextual Arabic explanations, ensuring students develop bilingual domain expertise.
- Features an AI-powered interactive chatbot that answers questions about the content, facilitating deeper comprehension and active learning.
- Generates concise summaries, keywords, and quizzes from videos, enhancing knowledge retention and engagement.

By democratizing access to world-class educational content, Dubljlly empowers Arabic-speaking students to overcome language barriers and access the same quality of education available to English speakers. The project specifically targets school and university students, researchers, and professionals in Jordan, with plans to expand to the broader Arab world, aligning with Jordan’s Digital Transformation Strategy [3]. Dubljlly stands at the intersection of educational technology and language accessibility, using AI to create a more equitable global learning ecosystem.

1.2 Purpose of this Document

This document serves as a comprehensive technical report detailing the development, implementation, and evaluation of the Dubljly AI-powered video dubbing platform. It fulfills multiple critical purposes as part of the graduation project requirements for the Bachelor of Information Technology in Computer Information Systems at Yarmouk University:

- **Academic Documentation:** Constitutes the primary deliverable, providing detailed documentation of the research methodology, technical implementation, and project outcomes.
- **Technical Reference:** Outlines the system architecture, component integration, API development, and decision-making process behind technology selections, serving as a guide for future developers.
- **Project Validation:** Validates the feasibility and effectiveness of the proposed solution through analysis of the first-phase implementation, demonstrating successful integration of speech-to-text, machine translation, and text-to-speech technologies.
- **Future Development Roadmap:** Establishes a foundation for future enhancements and scaling, providing insights to guide subsequent development phases.

1.3 Overview of this Document

This document is structured to provide a systematic progression from project introduction through technical implementation and validation, adhering to the Yarmouk University report template [2]. It is organized into five main chapters:

- **Chapter 1 - Introduction:** Establishes the project foundation by defining the problem statement, objectives, and document structure. It analyzes existing systems, identifying their limitations and contextualizing the need for Dubljly.
- **Chapter 2 - System Analysis:** Presents a comprehensive analysis of system requirements, including data flow diagrams, use-case diagrams, functional and non-functional requirements, class diagrams, proposed solutions, and alternatives.
- **Chapter 3 - Design Considerations:** Details design constraints, hardware and software environment, end-user characteristics, and architectural strategies employed in the project.
- **Chapter 4 - System Design:** Provides technical specifications of the system architecture, major modules (video acquisition, audio processing, LLM, TTS, video integration), and detailed component descriptions.
- **Chapter 5 - Implementation and Validation:** Documents the development process, integration details, testing methodologies, and performance validation of the Dubljly platform.

Appendices include code snippets (Appendix A) and references (Appendix References), ensuring a complete technical record for academic and future reference.

1.4 Existing System

1.4.1 Existing System Description

The current landscape of video dubbing solutions encompasses multiple categories, each with distinct technological approaches and target audiences [1]:

- **AI-Powered Video Dubbing Platforms:**
 - **Dubverse.ai:** Offers automated voice cloning and multilingual dubbing, supporting multiple languages. It focuses on commercial and marketing content, lacking educational-specific features like terminology preservation [8].
 - **Murf.ai:** Provides natural-sounding voiceovers with Arabic support, but is designed for corporate content, with limited academic context awareness [9].
- **Professional Dubbing Studios:**
 - **Traditional Studios** (e.g., SDI Media, Iyuno-SDI Group): Employ human voice actors and sound engineers, producing high-quality dubs. However, they require 4–8 weeks and cost \$2,000–\$8,000 per hour, making them inaccessible for educational purposes [1].
 - **Educational Specialists** (e.g., VSI London): Focus on academic content but are costly (\$3,000–\$10,000/hour) and slow (6–12 weeks) [1].
- **Video Platform Integrated Solutions:**
 - **YouTube Audio Track:** Allows manual upload of dubbed tracks but lacks automation [1].
 - **TikTok Voice Effects:** Offers basic translation for short-form content, unsuitable for educational videos [1].
- **Enterprise Solutions:**
 - **Rask.ai:** Targets corporate communications with voice cloning and lip-sync, but lacks educational optimization [10].
 - **Papercup:** Focuses on media and news, not academic content [11].

1.4.2 Problems in the Existing System

A systematic evaluation of existing solutions reveals critical deficiencies that justify the need for Dublily [1]:

- **Inadequate Educational Content Specialization:**
 - *Technical Terminology Mishandling:* Systems mistranslate terms like “algorithm” or “quantum mechanics,” causing confusion.

- *Context Loss*: Generic translations fail to maintain academic discourse, leading to misleading outputs.
- **Audio Quality and Synchronization Deficiencies:**
 - *Robotic Voices*: Synthesized audio lacks natural intonation, reducing engagement.
 - *Temporal Misalignment*: Poor audio-visual sync disrupts learning.
 - *Inconsistent Audio*: Varying quality affects the viewing experience.
- **Limited Arabic Language Optimization:**
 - *Linguistic Challenges*: Inadequate support for Arabic’s right-to-left text and morphology.
 - *Dialect Issues*: Lack of regional dialect support limits effectiveness.
- **Absence of Interactive Learning Enhancement:**
 - *Passive Consumption*: No tools for summarization or interactive Q&A.
 - *Non-Adaptive*: One-size-fits-all approach fails diverse learners.
- **Scalability and Economic Barriers:**
 - *High Costs*: Professional services are unaffordable for students and institutions.
 - *Resource-Intensive*: AI solutions require significant computational resources, scaling poorly.

Dubljly addresses these gaps through an education-focused, Arabic-optimized, cost-effective platform with interactive learning features.

2 System Analysis

3 Design Considerations

4 System Design

5 Implementation and Validation

A Code

B References

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